From Distributional to Overton Pluralism: Investigating Large Language Model Alignment

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Abstract

The alignment process changes several properties of a large language model's (LLM's) output distribution. We analyze two aspects of post-alignment distributional shift of LLM responses. First, we re-examine previously reported reductions in response diversity post-alignment. Our analysis suggests that an apparent drop in the diversity of responses is largely explained by quality control and information aggregation. Alignment suppresses irrelevant and unhelpful content while shifting the output distribution toward longer responses that cover information spanning several responses from the base LLM, essentially presenting diverse information in a single response. Finding little evidence that alignment suppresses useful information, it is natural to ask the opposite question: do aligned models surface information that cannot be recovered from base models? Our second investigation shows this is not the case and the behavior of aligned models is recoverable from base models without fine-tuning. A combination of in-context examples and lower-resolution semantic hints about response content can elicit responses from base LLMs that are as similar to alignment-tuned LLM responses as alignment-tuned LLM responses are to each other. Taken together, these results indicate that current alignment techniques capture but do not extend the useful subset of assistant-like base LLM behavior, providing further evidence for the Superficial Alignment Hypothesis. They also show that in-context alignment can go surprisingly far as a strategy for imitating aligned LLMs without fine-tuning. Our code and data is available at `this url`.

1 Introduction

After massive scale pre-training over trillions of tokens to learn the next token distribution, large language models (LLMs) go through an alignment step such as supervised fine-tuning (SFT) (Sanh et al., 2022), reinforcement learning from human feedback (RLHF) (Stiennon et al., 2020; Ouyang et al., 2022), or direct preference optimization (DPO) (Rafailov et al., 2024) which changes the conditional distribution defined by the LLM to favor responses that humans prefer. How does alignment change an LLM's distribution, beyond the intended consequence of improving the usefulness of outputs? Prior work has observed that it also causes significant reductions in response diversity (Khalifa et al., 2020; Casper et al., 2023; Go et al., 2023; Kirk et al., 2024). Reductions in diversity are concerning, given the broad demand for more personalized models to accommodate diverse human preferences (Jang et al., 2023; Cheng et al., 2023).

An orthogonal line of work has argued that fine-tuning is largely superficial. The Superficial Alignment Hypothesis conjectures that observed capabilities of aligned LLMs are learned primarily during pre-training and current techniques merely amplify an existing subdistribution corresponding to assistant-like responses (Zhou et al., 2024). Combined with observations about diversity, superficial alignment suggests a certain picture of model behavior: there are many types of responses from base models, and fine-tuning boosts the likelihood of returning certain responses without changing their content.

This paper provides empirical support for this characterization of current alignment techniques. We demonstrate that the output distribution of an alignment-tuned model corresponds to the useful output **subdistribution** of the corresponding base model in task-oriented settings. Our argument is divided into two sections which we describe below.

First, we ask whether a loss of diversity after alignment actually corresponds to a loss in diversity of the information content of those responses. Contrary to prior results (Bai et al., 2022; Kirk et al., 2024), our experiments suggest the answer is no. Figure 1 illustrates our findings. Samples

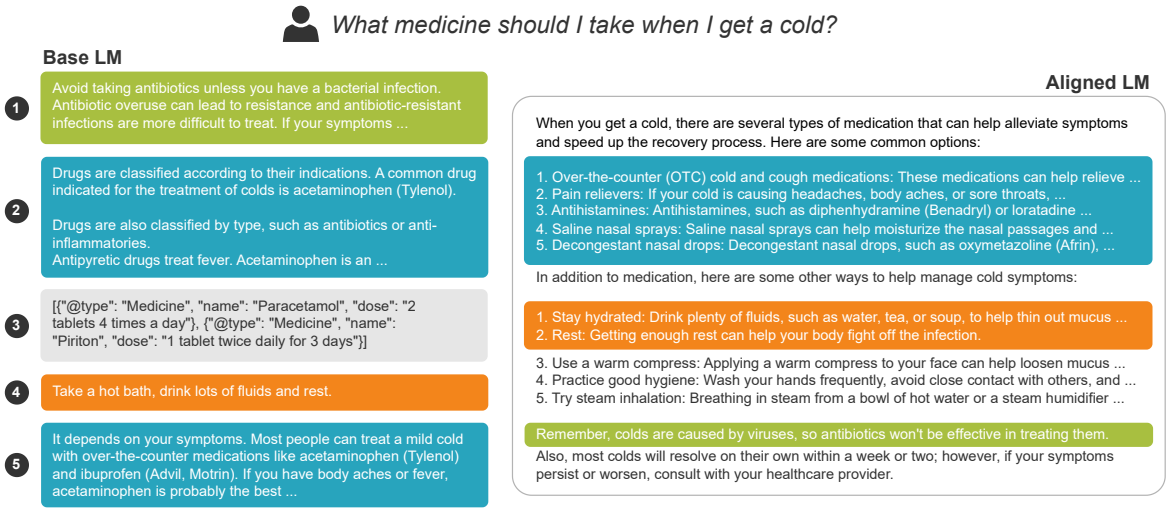

Figure 1: Comparing outputs from an unaligned (left) and aligned (right) language model pair. A single response from the aligned model contains useful information only surfaced by the unaligned model with repeated sampling while omitting unhelpful content.

from aligned models are often homogeneous and resemble the one shown here, whereas samples from base models (five shown) exhibit a broad range of semantics. However, some responses such as the JSON blob in gray (the third response) are not useful, and aligned models failing to cover these is acceptable. The increase in length from the response of the aligned model means that it actually covers many of the useful, distinct responses from the base language model *simultaneously*. We interpret this finding as a shift from distributional to Overton pluralism in the context of a recently proposed framework (Sorensen et al., 2024) and characterize this behavior on two datasets of questions with open-ended answers (Zhou et al., 2024; Wan et al., 2024) that arise due to points of scientific debate or differences in opinion.

If the aligned response is a filtered, aggregated view of base model responses, can this behavior be recovered from a base model without fine-tuning? Prior work has shown in-context learning (ICL) can elicit responses from a base model that are comparable to responses in aligned models on dimensions like helpfulness and factuality (Lin et al., 2024). We extend this study, improving in-context example selection and providing lexical hints that nudge models to produce content similar to aligned models. With our prompt guiding a base model to behave like an aligned model, the base model can produce samples strikingly similar to those from that aligned model, on both Llama 2 and Mistral. This suggests that the distribution of the base model *can* cover the responses from the aligned model even in the absence of parameter updates.

Together, our study provides an in-depth analysis on how alignment changes responses from LLMs, in terms of response diversity and prompting base models to produce responses that resemble responses from their aligned counterparts. Our success in imitating aligned behavior without fine-tuning suggests possibilities for rapid prototyping and personalization of LLMs.

2 Background

2.1 Goals of alignment

Alignment is the general process of shifting the conditional distribution defined by a pre-trained LLM away from the base model's distribution and toward useful task-oriented responses that align with user expectations and preferences. Helpfulness, factuality, and safety (Zou et al., 2023; Andriushchenko et al., 2024) are common objectives for the alignment process. Prior work (Amodei et al., 2016; Bai et al., 2022) measures whether such desired properties are achieved through alignment, which differs from our main goal of studying the distribution shifts over the semantics of answers for open-ended questions.

Diversity of responses is typically *not* an explicit goal of alignment. After alignment, model responses should be informative and cover different alternatives, and aligned models should be able to handle prompts corresponding to a range of possible situations. The ability to get diverse answers from an aligned model for the same prompt is present in APIs (e.g., the ability to change decoding parameters like sampling temperature, or

the presence of "creative" mode in tools like Bing Chat), but rarely an explicitly evaluated goal.

Alignment methods typically involve parameter updates, such as supervised fine-tuning (SFT) or reinforcement learning on human preferences (RLHF). In this work we will experiment with *in-context alignment* (Lin et al., 2024), specifically the URIAL method, which crafts a few-shot prompt enabling a base LLM to generate responses resembling those from their aligned counterparts. The ability to do this without updating parameters supports the Superficial Alignment Hypothesis.

2.2 Measuring alignment's effects with open-ended QA datasets

To measure the semantic diversity of LLMs' outputs before and after alignment, we chose two datasets containing open-ended questions where a range of valid answers from multiple perspectives exists. We aim to measure if alignment-tuning significantly alters the information surfaced in responses by reducing diversity or suppressing certain viewpoints. Summary statistics and examples are provided in Table 1.

CONFLICTINGQA (Wan et al., 2024) consists of short questions on controversial topics. Although each question can technically be answered with a simple yes or no, high-quality responses provide additional context and elaboration, particularly as the answer may be "it depends".

LIMA-OE is a subset of the LIMA dataset, which is selected from community Q&A forums such as Reddit and written by the authors of prior work (Zhou et al., 2024). We further manually filter this dataset to only have questions that can have multiple valid answers, and refer to this subset as LIMA-OE (for "open-ended"). Questions in LIMA-OE consist of a mix of information-seeking behaviors such as broad queries and requests for suggestions or advice. Excluded questions primarily relate to coding, document (re)writing, and summarization tasks, which either have narrow solutions or exhibit quality notions like code efficiency that are not congruent with our evaluation.

Models We compare three variants of Llama 2 that reflect different dimensions of alignment. Llama 2 (7B) is a base language model, which we prompt zero-shot to answer the question, and Llama 2 Chat (7B) (Touvron et al., 2023) is its corresponding aligned model, which we prompt in the same way. We also compare against a Llama 2 model with in-context alignment via URIAL (Lin et al., 2024). In the remainder, we refer to these models as Llama 2 Base, Llama 2 Chat, and Llama 2 ICL respectively. In later sections, we also experiment with the Mistral family of models (Jiang et al., 2023) which we refer to analogously as Mistral Base and Mistral Instruct. We also include results for GPT-3.5 Turbo. Hyper-parameter settings and prompts are in Appendix C.

Dataset	Size	Length
CONFLICTINGQA	434	8
Is the Gender Wage Gap a Myth? Can you inherit genes for talent and skill? Were there dinosaurs on Noah's Ark?		
LIMA-OE	50	19
Why can parrots talk? Who is the greatest woman in history? How can I improve my time management skills?		

Table 1: Datasets used in this paper. Length gives the average number of words per question.

2.3 Observable effects of alignment

Previous work has noted tradeoffs between quality and diversity of alignment (Florian et al., 2024), particularly on tasks like summarization (Kirk et al., 2024). In this section, we validate these results on two datasets we study.

Evaluation criteria We measure **response quality** along five dimensions using GPT-4 (`gpt-4-1106-preview`) and the Just-Eval prompt from URIAL, which was previously found to yield high agreement rates between human preferences in closely related settings (Lin et al., 2024). Each response is rated on each aspect from a 1 to 5 scale, corresponding to strong disagreement to strong agreement about whether the response exhibits that aspect. While LLM evaluators have weaknesses and various biases, documented biases (Stureborg et al., 2024) such as anchoring bias and failure to use the entire score range don't bias our results, as we are comparing relative rankings of similar systems with a fixed protocol. We additionally performed a small-scale qualitative review of LLM judgments and found we generally agreed with the relative ordering of responses.

We measure **response similarity** between multiple samples using lexical measures (Li et al., 2016; Zhu et al., 2018; Liu et al., 2022) which correlate with human judgments of diversity (Tevet and Berant, 2020; Shaib et al., 2024). Given a set of sampled responses to a prompt, we measure the

Model	Prompt	Quality (measured by GPT-4)					Output	
		Helpfulness	Clarity	Factuality	Depth	Engagement	Self-Sim	Length
ConflictingQA								
Llama 2	-	2.21	3.29	2.30	1.49	1.78	0.17	37
Llama 2	URIAL	4.00	4.72	3.64	3.18	3.79	0.23	203
Llama 2 Chat	-	4.78	4.94	4.26	4.48	3.88	0.36	412
GPT-3.5 Turbo	-	4.37	4.95	4.65	3.08	3.10	0.45	98
LIMA-OE								
Llama 2	-	3.08	3.80	3.43	2.36	2.54	0.11	149
Llama 2	URIAL	4.34	4.88	4.56	3.46	3.86	0.24	261
Llama 2 Chat	-	4.94	5.00	4.62	4.72	4.30	0.34	552
GPT-3.5 Turbo	-	4.90	5.00	4.92	4.48	4.14	0.36	355

Table 2: Quality and diversity metrics evaluated on CONFLICTINGQA and LIMA-OE. In line with previous observations, alignment increases response quality and length while reducing diversity (increasing the similarity between sampled responses). All Llama 2 models are 7B variants.

per-input similarity of responses by averaging Jaccard similarity over all pairs after converting each response into a set of unique word stems. We refer to this as **Self-Sim** and display the average over for N = 5 samples for each input.

Results Our results are summarized in Table 2. In-context alignment with the URIAL prompt leads to a performance boost of the Llama 2 base model, closing the gap towards its aligned counterpart, Llama 2 Chat. Llama 2 Chat matches or exceeds the performance of GPT-3.5 Turbo according to all metrics except factuality. It also sees a large increase in self-similarity compared to the Llama 2 base model. While URIAL prompting clearly improves the quality of base model response, the effect is less pronounced than in fine-tuned models; accordingly, its responses are more self-similar than base model responses but less self-similar than chat-tuned responses. Alignment also increases the length of responses, an observation previously reported in the literature (Stiennon et al., 2020; Singhal et al., 2023).

These results corroborate findings of reduced diversity post-alignment (Casper et al., 2023; Ryan et al., 2024). In the following section, we seek to investigate more precisely how diversity changes post-alignment and whether informative responses from the base model are suppressed or lost.

3 Decreased diversity post-alignment

We now investigate how the diversity of output changes during the alignment process. Reducing diversity of responses can be part of improving response quality: for instance, when prompted to solve a math problem, the ideal language model returns the single correct answer and not a "diverse" range of incorrect answers. However, this is not as clearly true on the datasets we investigate.

To ground our discussion, we appeal to a recently proposed framework for the closely related concept of pluralism, which defines several distinct ways for LLMs to be pluralistic, or "capable of representing a diverse set of human values and perspectives" (Sorensen et al., 2024). Their notion of *distributional pluralism*, or the ability of LLM to surface diverse information across samples, can be reflected by the lexical diversity metric we introduced (**Self-Sim**). On the other hand, *Overton pluralism*, which requires surfacing diverse information within a single response, is not captured this way, but may be captured by length, as capturing diverse information can lead to longer answers. We observe that aligned model outputs tend to be longer, leading us to question if the alignment process might lead LLMs to exhibit different types of diversity (from pluralistic to Overton).

In this section, we compare the outputs from the base and aligned models to measure how answer semantics change after the alignment process, specifically by looking at information "missing" in the response post-alignment. **Our main hypothesis is that the only information missing from the response is information that is not useful (i.e., low-quality).**

3.1 Helpfulness and missingness

Setting We aim to evaluate the helpfulness of information present in the base model responses, $\mathbf{y}$, that is missing in the aligned model responses, $\mathbf{y}_r$. At a high level, we measure coverage between two responses. If coverage is *high*, then no information is missing. We measure this with two metrics: